2023 SSMO Relay Round 1

SMO Team

RR 1 Part 1: Compute the remainder when $2022^{2021^{2020^{\cdots}}}$ is divided by 2023.

RR 1 Part 2: Let $T = TNYWR$. Let $a_0 = 3$, $a_1 = 1$, $a_2 = T$, and let $a_n = a_{n-1} - \frac{a_{n-3}}{8}$ for $n \geq 3$. Find

$$\sum_{i=0}^{\infty} a_i.$$

RR 1 Part 3: Let $T = TNYWR$. Find the number of solutions to the equation

$$\sec^T(Tx) - \tan^T(Tx) = 1$$

such that $0 \leq x \leq \pi$.

2023 SSMO Relay Round 2

SMO Team

RR 2 Part 1: Consider the cubic polynomial $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c, and d are single-digit integers, which has roots of approximately

$$x \approx -0.9518399, 0.2055095, 1.460616.$$

Compute $|f(3)|$.

RR 2 Part 2: Let $T = TNYWR$. Suppose that $L = \left\lfloor \sqrt{T} \right\rfloor$ points are evenly spaced around a circle. Find the number of ways to select $k \geq 3$ points such that the k-gon formed strictly contains the center of the circle.

RR 2 Part 3: Let $T = TNYWR$. In a committee of 2023 people, T are scientists and the rest are builders. In order to make a building, $\frac{T}{2}$ people must be chosen with at least one scientist and one builder. Find the largest nonnegative integer b such that 2^b divides the number of ways to do this.

2023 SSMO Relay Round 3

SMO Team

RR 3 Part 1: In triangle ABC with $AB = 13$, $AC = 14$, and $BC = 15$, circles ω_1, ω_2, and ω_3 are drawn, centered at A, B, and C, respectively. Each of the three circles is externally tangent to the two other circles. The radius of a circle ω such that ω is externally tangent to ω_1, ω_2, and ω_3 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

RR 3 Part 2: Let $T = TNYWR$. A spray painter paints all points in the coordinate plane within Euclidean distance 2 of their path. The painter walks along the perimeters of concentric squares centered at the origin, starting with the outermost square and moving inward to the innermost square:

- For each integer t from T down to 1, the painter walks along the square with vertices $(\pm t, \pm t)$.
- The spray color alternates based on t: the painter sprays blue if t is even and red if t is odd.
- If a region is painted multiple times, only the most recently applied color is visible.

The positive difference between the visible area of the blue region and the visible area of the red region can be written as $a + b\pi$, where a and b are integers. Find the remainder when $a + b$ is divided by 100.

RR 3 Part 3: Let $T = TNYWR$. A town is laid out as a $T \times T$ grid of unit squares with $T + 1$ vertical lines and $T + 1$ horizontal lines, forming $2T^2 + 2T$ unit-length segments. A snowplow can start and end anywhere and must clear every segment at least once. What is the minimum number of units the snowplow must drive?

2023 SSMO Relay Round 4

SMO Team

RR 4 Part 1: Five numbers are chosen independently and uniformly at random from the set $\{1, 2, 3, 4, 5\}$, with replacement. The probability that the median of these 5 numbers is greater than 3 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

RR 4 Part 2: Let $T = TNYWR$. Martha starts out with the number 7 on a calculator with three buttons that multiply the current display by $\frac{1}{2}$, $\frac{2}{3}$, and $\frac{5}{6}$, respectively. She performs T rounds of operations. In each round, she chooses one of the three buttons independently and uniformly at random and presses it three times in a row. The expected value of the final value shown on her calculator can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the remainder when $m + n$ is divided by 100.

RR 4 Part 3: Let $T = TNYWR$. Suppose that $\left\lfloor \sqrt{T} \right\rfloor$ points are chosen on the sides of a triangle with area 1, where each point independently chooses one of the three sides uniformly at random and then a location on that side uniformly at random. Given that there is at least one point on each side, the expected value of the area of the polygon formed by connecting the chosen points in counterclockwise order can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

2023 SSMO Relay Round 5

SMO Team

RR 5 Part 1: Let S_n be the set of all rational numbers of the form $0.\overline{a_1a_2a_3\ldots a_n}$, where $a_1, a_2, \ldots, a_n$ are nonzero digits. Find the value of

$$5\left(\sum_{n=1}^{\infty}\left(\sum_{a\in S_n}\frac{a}{10^n}\right)\right).$$

RR 5 Part 2: Let $T = TNYWR$. Compute

$$\left(\sum_{i=1}^{T}\text{lcm}\{1,2,\ldots,i\}\right) \pmod{720}.$$

RR 5 Part 3: Let $T = TNYWR$. Distinct integers $a, b \in \{1, 2, \ldots, T\}$ are chosen uniformly at random. Consider the maximum of $ax + by$ over all reals x and y that satisfy $x^2 + y^2 = T$. Find the remainder when the expected value of the square of this maximum is divided by 100.